arXiv:2305.17493v3 [cs.LG] 14 Apr 2024

THE CURSE OF RECURSION: TRAINING ON GENERATED DATA MAKES MODELS FORGET

Ilia Shumailov* University of Oxford
Zakhar Shumaylov* University of Cambridge
Yiren Zhao Imperial College London
Yarin Gal University of Oxford
Nicolas Papernot University of Toronto & Vector Institute
Ross Anderson University of Cambridge & University of Edinburgh

ABSTRACT

Stable Diffusion revolutionised image creation from descriptive text. GPT-2, GPT-3(.5) and GPT-4 demonstrated astonishing performance across a variety of language tasks. ChatGPT introduced such language models to the general public. It is now clear that large language models (LLMs) are here to stay, and will bring about drastic change in the whole ecosystem of online text and images. In this paper we consider what the future might hold. What will happen to GPT-$\{n\}$ once LLMs contribute much of the language found online? We find that use of model-generated content in training causes irreversible defects in the resulting models, where tails of the original content distribution disappear. We refer to this effect as *model collapse*[1] and show that it can occur in Variational Autoencoders, Gaussian Mixture Models and LLMs. We build theoretical intuition behind the phenomenon and portray its ubiquity amongst all learned generative models. We demonstrate that it has to be taken seriously if we are to sustain the benefits of training from large-scale data scraped from the web. Indeed, the value of data collected about genuine human interactions with systems will be increasingly valuable in the presence of content generated by LLMs in data crawled from the Internet.

1 Introduction

A lot of human communication happens online. Billions of emails are exchanged daily, along with billions of social-media messages and millions of news articles. Almost all of this material was produced and curated only by humans in the early years of the worldwide web, yet since the turn of the century search engines have come to determine what people can find, and in the past decade smart text editors with spelling and grammar correction have helped tweak what we produce. Now, text can not only be groomed and analysed efficiently; it can also be generated – by large language models (LLMs). These models now (arguably) pass a weaker form of the Turing test in the sense that their output cannot be reliably distinguished from text written by humans [Solaiman et al., 2019].

The development of LLMs is quite involved and requires masses of training data. Anecdotally, some powerful recent models are trained using scrapes of much of the Internet, then further fine-tuned with reinforcement learning from human feedback (RLHF) [Griffith et al., 2013, OpenAI, 2023]. This further boosts the effective dataset size. Yet while current LLMs [Devlin et al., 2018, Liu et al., 2019, Brown et al., 2020, Zhang et al., 2022], including GPT-4, were trained on predominantly human-generated text, this may change in the future. If most future models' training data is also scraped from the web, then they will inevitably come to train on data produced by their predecessors. In this paper, we investigate what happens when text produced, *e.g.* by a version of GPT, forms most of the training dataset of following models. What happens to GPT versions GPT-$\{n\}$ as generation n increases?[2]

[1]The name is inspired by the Generative Adversarial Networks (GAN) literature on mode collapse, where GANs start producing a limited set of outputs that all trick the discriminator. *Model Collapse* is a process whereby models eventually converge to a state similar to that of a GAN Mode Collapse. The original version of this paper referred to this effect as 'model dementia', but we decided to change this following feedback that it trivialised the medical notion of 'dementia' and could cause offence.

[2]This is not limited to text models; one can also consider what happens when music created by human composers and played by human musicians trains models whose output trains other models.

We discover that learning from data produced by other models causes *model collapse* – a degenerative process whereby, over time, models forget the true underlying data distribution, even in the absence of a shift in the distribution over time. We give examples of *model collapse* for Gaussian Mixture Models (GMMs), Variational Autoencoders (VAE) and Large Language models (LLMs). We show that over time we start losing information about the true distribution, which first starts with tails disappearing, and over the generations learned behaviours start converging to a point estimate with very small variance. Furthermore, we show that this process is inevitable, even for cases with almost ideal conditions for long-term learning *i.e.* no function estimation error.

Finally, we discuss the broader implications of *model collapse*. We note that access to the original data distribution is crucial: in learning where the tails of the underlying distribution matter, one needs access to real human-produced data. In other words, the use of LLMs at scale to publish content on the Internet will pollute the collection of data to train them: data about human interactions with LLMs will be increasingly valuable.

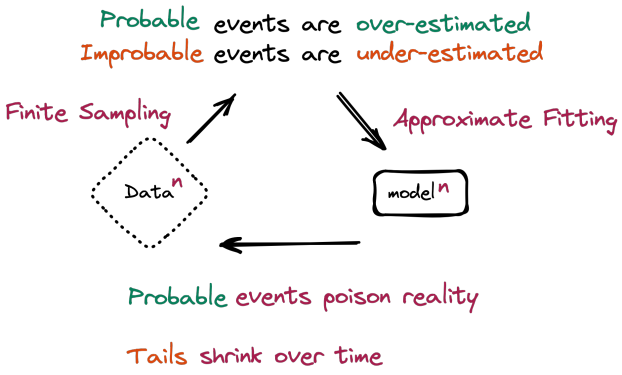

Figure 1: *Model Collapse* refers to a degenerative learning process where models start forgetting improbable events over time, as the model becomes poisoned with its own projection of reality.

This paper is structured as follows. First, in Sections 3 and 4 we describe the reasons why *model collapse* happens. To best describe the intuition, we present a simple example of a single-dimensional Gaussian where errors due to sampling inevitably cause *model collapse*, which are then extended to a multidimensional generative model under some assumptions. Under both models, similar lower bounds are derived on the risk, defined in terms of the Wasserstein distance from the true distribution. Next, we turn to GMMs and VAEs to show that additional functional approximation errors further exacerbate *model collapse*. Finally, we discuss the most commonly used setting of fine-tuned language models, where we report that only early signs of *model collapse* can be detected, if models are fine-tuned as opposed to trained from scratch.

In this paper we make the following contributions:

- We demonstrate the existence of a degenerative process in learning and name it *model collapse*;
- We demonstrate that *model collapse* exists in a variety of different model types and datasets;
- We show that, to avoid *model collapse*, access to genuine human-generated content is essential.

2 Related work

In this section we are going to cover two closest concepts to *model collapse* from existing literature: catastrophic forgetting and data poisoning. Neither is able to explain the phenomenon of *model collapse* fully, as the setting is fundamentally different, but they provide another perspective on the observed phenomenon.

2.1 Continual learning and catastrophic forgetting

Unlike traditional machine learning which seeks to learn from a static data distribution, *continual learning* attempts to learn from a dynamic one, where data are supplied in a sequential fashion [Van de Ven and Tolias, 2019]. This tends to be task-based, where data are provided with delineated task boundaries; *e.g.*, classifying dogs from cats and recognising handwritten digits. Our work is more similar to task-free continual learning [Aljundi et al., 2019] where data distributions gradually change without the notion of separate tasks. Our work examines a particular scenario in which the changed data distributions arise from the model itself, as a result of training in the previous iteration.

Example of text outputs of an OPT-125m model affected by *Model Collapse*– models degrade over generations, where each new generation is trained on data produced by the previous generation.

Input: some started before 1360 — was typically accomplished by a master mason and a small team of itinerant masons, supplemented by local parish labourers, according to Poyntz Wright. But other authors reject this model, suggesting instead that leading architects designed the parish church towers based on early examples of Perpendicular

Outputs:

Gen 0: Revival architecture such as St. John's Cathedral in London. The earliest surviving example of Perpendicular Revival architecture is found in the 18th @-@ century Church of Our Lady of Guernsey, which dates from the late 19th century. There are two types of perpendicular churches : those

Gen 1: architecture such as St. Peter's Basilica in Rome or St. Peter's Basilica in Buenos Aires. There is no evidence that any of these buildings were built during the reign of Pope Innocent III, but it is possible that they may have been built during the reign of his successor, Pope Innocent

Gen 7: architecture in England. In an interview with The New York Times, Wright said : " I don 't think there is anything wrong with me being able to do what I want to do. It just doesn 't work for me. " He added : " I don 't know if you can call it funny,

Gen 9: architecture. In addition to being home to some of the world's largest populations of black @-@ tailed jackrabbits, white @-@ tailed jackrabbits, blue @-@ tailed jackrabbits, red @-@ tailed jackrabbits, yellow @-

A typical challenge in continual learning is that the model forgets previous samples when learning new information; this is known as *catastrophic forgetting* [Kirkpatrick et al., 2017]. A typical way of preventing it is to use regularisations (Memory Aware Synpass [Aljundi et al., 2018]) or just rely on data (*e.g.* Learning without Forgetting [Li and Hoiem, 2017]). This has an indirect connection to our work, yet differs since the data in the process of *model collapse* are generated by different generations of models.

2.2 Data poisoning

Poisoning attacks are crafted and inserted during training in order to degrade the model's performance when deployed [Biggio et al., 2012]. Malicious data can be inserted into training data to induce unintended behaviors that can be activated by special triggers [Gu et al., 2017]. The early literature on data poisoning focused mainly on supervised learning, where classifiers are trained with labeled samples. But with the emergence of contrastive learning [Radford et al., 2021] and LLMs [Brown et al., 2020], more recent models are trained with large-scale web crawls, making data poisoning attacks more feasible on these untrustworthy web sources. Recent studies have demonstrated that web-scale datasets can be poisoned by introducing malicious data into a small percentage of samples [Carlini and Terzis, 2021, Carlini et al., 2023].

3 What is *Model Collapse*?

Definition 3.1 (*Model Collapse*)**.** *Model Collapse* is a degenerative process affecting generations of learned generative models, where generated data end up polluting the training set of the next generation of models; being trained on polluted data, they then mis-perceive reality. We separate two special cases: **early *model collapse*** and **late *model collapse***. In early *model collapse* the model begins losing information about the tails of the distribution; in the late *model collapse* model entangles different modes of the original distributions and converges to a distribution that carries little resemblance to the original one, often with very small variance.

Note that this process is different from the process of *catastrophic forgetting* in that we are considering multiple models over time, in which our models do not forget previously learned data, but rather start misinterpreting what they believe to be real, by reinforcing their own beliefs.

This process occurs due to two specific sources of error compounding over generations and causing deviation from the original model. Of these, one source of error plays a primary role, and in the absence of it, the process would not occur beyond the first generation.